Group Project Descriptor: Developing an Intelligent Line-Following Algorithm for the Quanser Qbot

Course: ENG5337 Advanced Artificial Intelligence

The objective of this project is to apply advanced AI techniques learned during the course to develop a robust line-following algorithm for the Quanser Qbot. In groups, students will develop their chosen framework in Python and test it in the simulated Quanser warehouse environment.

Project Description

Students will work in groups to select an AI framework and use it to create a line-following algorithm for the Quanser Qbot. The project will be divided into the following phases:

1. Data Selection and Model Identification

- **Analyse the key characteristics of the line-following task.** Based on this analysis, decide what input data you will use (e.g., sensor readings) and how you will filter, process, or transform it. If your chosen approach requires labelled data, determine how you will assign labels. Incorporate any feedback from your earlier individual project submission.
- **Select and justify a suitable machine learning approach.** You may choose from models introduced in the coursework (such as neural networks or reinforcement learning) or explore alternative algorithms. When making your decision, consider factors such as:
    - The nature of the problem
    - Dataset size and quality
    - Computational resources
    - Model complexity and interpretability

2. Algorithm Development

- **Develop and implement your chosen machine learning model** using either a neural network (e.g., with PyTorch), a reinforcement learning algorithm, or another suitable method, based on the data or environment prepared in Step 1.
- When designing your model, consider:

    - **Core components:** choice of cost/reward function, optimiser, and any regularisation techniques.
    - **Hyperparameter tuning:** such as learning rate, batch size, activation functions, number of layers/neurons (for NNs), or exploration strategies and discount factors (for RL).
- **Evaluate the performance of your model** using appropriate metrics (e.g., accuracy, F1-score for supervised models; cumulative reward or success rate for RL).

Important: Once you have collected your data in the Quanser software, develop and test your model outside the Quanser environment. When satisfied with its performance, save the model file and load it within Quanser for deployment.

3. Simulation Testing

- Test the model from Step 2 in the Quanser simulated environment (use the warehouse environment from tutorials).
- If the QBot does not successfully follow the line, revisit Steps 1 and 2 to refine your strategy.

4. Benchmarking

- In your individual project, you each proposed a minimum of two metrics to compare your ML-driven line-follower with classic PID control. Implement at least three metrics to **compare your ML-driven line-following approach (NN, RL, or other) with the classic control algorithm** provided in the Quanser tutorials.

Deliverables

This project will be evaluated in an oral group presentation:

- 5 minute demo: The group will be asked to demo their line follower live in the simulated Quanser Interactive Labs environment. Note you may be asked by markers to manually move the robot to a designated location and follow a specific line.
- 15 minute presentation: The group will then give a 15 minute presentation outlining their methodology, testing and refinement process, results and benchmarking performance. **The markers will be looking for justification behind each decision made in the design of the methodology**. The group should also include suggestions for refining and improving their presented framework.
- 10 minute Q&A. The panel will ask questions to clarify presentation content and explore the group's understanding. Questions will also be taken from your peers.

Groups must submit their slides by 5pm on Wednesday 25th March 2026. You should also submit a zip file with the necessary Quanser scripts required to run the line-follower (don't forget the qbot_platform_functions.py file), the ML- model file, and the python script in which the model was trained, validated and tested.

Evaluation Criteria

Grade Range (Highest to Lowest)	**A1, A2, A3, A4, A5**	**B1, B2, B3**	**C1, C2, C3**	**D1, D2, D3**	**E1, E2, E3**	**F1, F2, F3**	**G1, G2, H**
Descriptor	**Excellent**	**Very Good**	**Good**	**Satisfactory**	**Weak**	**Poor**	**G: Very Poor** **H: No Attainment**
Delivery *(Weighting = 1)*	*Confident, clear, and unhesitating delivery. Held attention of audience. Easy to follow arguments.*	*Was confident but perhaps a few minor flaws (such as hesitation, talking too fast etc).*	*Perhaps slightly lacking in confidence or possibly not speaking quite clearly enough.*	*Overall a reasonable delivery, but there were issues regarding clarity, or fluency.*	*A hesitant or unclear delivery made understanding the presentation difficult.*	*Hesitant, unclear, monotonous, hard to maintain attention. Difficult to follow arguments.*	*No fluency or clarity. Too many basic errors, e.g. mumbling or talking to screen.*
Slides *(Weighting = 1)*	*Exceptionally clear slides. Simple design, large enough font, not too much material on slides. A professional quality presentation.*	*Clear slides but perhaps the occasional flaw (font size, colour scheme etc), but overall an impressive presentation.*	*There may be a number of errors, on the slides but overall, still clear and flaws do not detract significantly from content.*	*Consistent errors on many slides but not of a significant nature. A reasonable effort but flaws have detracted from presentation.*	*Significantly flawed slides. Basic errors such as small font size, too much content on slides, over-elaborate design.*	*Not only are slides poor, but they make it difficult to follow argument.*	*Very poor slides, basic errors on every slide. Impossible to follow the technical arguments.*
Technical Content *(Weighting = 2)*	*There is a well-judged amount of high-level technical content in the presentation giving an excellent account of a challenging technical task.*	*The presentation has a very good level of technical content, clearly expressed, with only a small amount of superfluous information.*	*Overall, the content is sufficient to give the audience a good account of the technical work undertaken.*	*There is some irrelevant non-pertinent material, but overall, the technical content is satisfactory.*	*The presentation has only limited technical content with too much general background information.*	*The technical content is low in terms of level and quantity.*	*Little or no relevant technical content evident.*
Structure *(Weighting = 1)*	*Structure of the presentation makes understanding the technical arguments exceptionally clear.*	*A very well-structured presentation with everything where it should be to provide clarity.*	*Overall a well-structured presentation but perhaps one or two slides are misplaced.*	*Some elements of the presentation are not clear as the structure is slightly confused.*	*A badly structured presentation giving a confused picture of the project making it difficult to follow the arguments.*	*Although there is some structure to the presentation it is very confused, and it is almost impossible to follow.*	*No discernible attempt at a logical structure.*
Response to Questions *(Weighting = 2)*	*Answered all questions clearly and confidently. Gave the impression of having an excellent grasp of the subject.*	*Answered all questions competently. Has clearly developed a very good understanding of the subject.*	*Answered most questions well enough to conclude that the student has a developed a good understanding of the subject.*	*Gave some good answers but also some poor ones. Evidence of reasonable understanding of the subject.*	*Answered the majority of the questions poorly suggesting a lack of knowledge in the subject.*	*Gave some superficial answers but appears to have very little understanding of the subject.*	*Unable to give any sort of competent answer to any question.*